

Original article

Multimodal lightweight neural network for Alzheimer's disease diagnosis integrating neuroimaging and cognitive scores

Bhoomi Gupta^{a,*}, Ganesh Kanna Jegannathan^b, Mohammad Shabbir Alam^c,
Kottala Sri Yogi^{d,e}, Janjhyam Venkata Naga Ramesh^{f,g}, Vemula Jasmine Sowmya^h,
Isa Bayhanⁱ

^a Department of ITE, Maharaja Agrasen Institute of Technology, Delhi, India

^b Department of Information Technology, Maharaja Agrasen Institute of Technology, Delhi, India

^c Department of Computer Science, College of Engineering and Computer Science, Jazan University, Jazan, Kingdom of Saudi Arabia

^d Research Dept of Operations, Symbiosis Institute of Business Management, Hyderabad, Telangana, India

^e A Constituent College of Symbiosis International University Pune, India

^f Department of CSE, Graphic Era Hill University, Dehradun, 248002, India

^g Department of CSE, Graphic Era Deemed To Be University, Dehradun, 248002, Uttarakhand, India

^h Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, Vaddeswaram, Guntur Dist., Andhra Pradesh, 522302, India

ⁱ Bolu Abant Izzet Baysal University, 14030, Golkoy Campus, Bolu, Turkey

ARTICLE INFO

Article history:

Received 15 May 2025

Received in revised form 22 June 2025

Accepted 4 July 2025

Keywords:

Alzheimer's disease

Multimodal diagnosis

MRI

PET

Neuropsychological scores

Neural networks

Lightweight model

Feature fusion

ADNI

Transfer learning

ABSTRACT

Conventional single-modal approaches for auxiliary diagnosis of Alzheimer's disease (AD) face several limitations, including insufficient availability of expertly annotated imaging datasets, unstable feature extraction, and high computational demands. To address these challenges, we propose Light-Mo-DAD, a lightweight multimodal diagnostic neural network designed to integrate MRI, PET imaging, and neuropsychological assessment scores for enhanced AD detection. In the neuroimaging feature extraction module, redundancy-reduced convolutional operations are employed to capture fine-grained local features, while a global filtering mechanism enables the extraction of holistic spatial patterns. Multimodal feature fusion is achieved through spatial image registration and summation, allowing for effective integration of structural and functional imaging modalities. The neurocognitive feature extraction module utilizes depthwise separable convolutions to process cognitive assessment data, which are then fused with multimodal imaging features. To further enhance the model's discriminative capacity, transfer learning techniques are applied. A multilayer perceptron (MLP) classifier is incorporated to capture complex feature interactions and improve diagnostic precision. Evaluation on the ADNI dataset demonstrates that Light-Mo-DAD achieves 98.0% accuracy, 98.5% sensitivity, and 97.5% specificity, highlighting its robustness in early AD detection. These results suggest that the proposed architecture not only enhances diagnostic accuracy but also offers strong potential for real-time, mobile deployment in clinical settings, supporting neurologists in efficient and reliable Alzheimer's diagnosis.

© 2025 The Author(s). Published by Elsevier Masson SAS. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

1. Introduction

Alzheimer's disease (AD) is a multifaceted and irreversible neurological condition that predominantly impacts elderly individuals, resulting in progressive cognitive and functional deterioration. It is

the predominant kind of dementia, representing 60–80% of global dementia cases. Neuropathologically, Alzheimer's disease is marked by the aggregation of amyloid-beta plaques and neurofibrillary tangles formed by hyperphosphorylated tau protein, leading to synaptic impairment and neuronal degeneration [1]. Clinically, individuals demonstrate progressive memory deterioration, compromised reasoning abilities, and disorientation, frequently advancing to total reliance. Notwithstanding considerable progress in neuroscience, the prompt and precise diagnosis of Alzheimer's disease continues to pose difficulties owing to symptom overlap with other cognitive disorders and the intricacies of the illness's pathophysiology [2]. Thus, the amalgamation of neuroimaging, cognitive evaluations,

* Corresponding author.

E-mail addresses: bhoomigupta@mit.ac.in (B. Gupta),

Ganeshkanna22@gmail.com (G.K. Jegannathan), amushabbir@gmail.com

(M.S. Alam), dr.sriyogi@outlook.com (K.S. Yogi), jvnramesh@gmail.com

(J.V.N. Ramesh), vemulajasmine@gmail.com (V.J. Sowmya), isa.bayhan@gmail.com

(I. Bayhan).

and biomarker analysis has become crucial for enhancing diagnostic precision and comprehending disease advancement.

The rapid advancement of artificial intelligence and its integration with neuroscience has led to the development of numerous computational methods for the early detection of Alzheimer's disease. These methodologies are typically classified into two categories: traditional machine learning techniques and deep learning models. Traditional methods have utilized features extracted from MRI and PET images via region-of-interest analysis, employing classifiers such as multifold Bayesian kernelization (MBK) [3], support vector machines (SVM) [4], and linear discriminant analysis (LDA) [5] on multimodal data that includes neuroimaging [6], genetic information [7], and cerebrospinal fluid biomarkers. While these methods may effectively amalgamate numerous data sources, they often struggle to identify subtle clinical variations due to limited feature representation. The ADNI dataset shows notable gains in classification ability, especially when it comes to differentiating between patients with moderate cognitive impairment, those with AD, and cognitively normal people. By combining the structural, metabolic, and cognitive aspects of AD, Light-Mo-DAD offers a low-cost, workable approach to early diagnosis that is consistent with current neuroscientific knowledge of the disease's course. Deep learning approaches, on the other hand, offer superior capabilities for learning hierarchical features directly from raw data. For instance, some models employ 3D convolutional neural networks (CNNs) for comprehensive spatial feature extraction [8], while others use slice-wise convolutional networks in multiple planes combined with recurrent networks for classification [9]. Further, multimodal fusion using 3D CNNs and fully connected layers has demonstrated strong predictive performance [10,11]. However, these models often demand significant computational resources, making them less suitable for real-time or portable clinical applications. To address these challenges, some researchers have shifted toward 2D representations and lightweight architectures. Approaches such as PCA-Net with Broad Learning Systems (BLS) [12] and VGG-16-based models with preprocessed 2D slices [13] help reduce computational complexity, though often at the expense of spatial and contextual information critical for AD diagnosis. One prevalent non-motor symptom of Parkinson's disease (PD) that has a substantial impact on patients' quality of life and presents difficulties for clinical care is cognitive impairment [14]. Electroencephalography (EEG) is a commonly used technique for diagnosing epileptic episodes. It is exceedingly challenging and time-consuming to visually evaluate nonlinear EEG signals [15]. With the use of sophisticated machine learning algorithms, generative AI (GenAI) chatbots are becoming revolutionary instruments for improving patient education, information sharing, and counseling (EIC) in the medical field [16]. Reducing newborn mortality rates requires early asphyxia detection. While early detection with artificial intelligence (AI) can speed up the procedure and increase survival rates, traditional medical diagnosis techniques can be time-consuming [17]. To overcome these limitations, recent research has explored hybrid and efficient architectures. For example, [18] proposed models that integrate multi-scale feature extraction with genome-wide association analysis, as well as methods combining region-of-interest and genetic data features using ensemble learning. Additionally, lightweight networks that combine convolutional neural networks with Transformer modules, such as the model proposed by Mehta [19], have shown the ability to capture both local and global features, outperforming traditional CNNs and vision Transformer models across various neuroscience and medical imaging tasks.

In the field of AD-assisted diagnosis, single-modality data is difficult to provide doctors with comprehensive pathological information about patients. Although using a 3D convolutional neural network as a training method can achieve better accuracy, it in-

volves many training parameters and high device requirements. This study presents a novel and efficient framework for the auxiliary diagnosis of Alzheimer's Disease (AD), integrating multimodal neuroimaging and cognitive assessment scores. The main contributions are summarized as follows:

- 1. Design of a Lightweight Multimodal Diagnostic Network (Light-Mo-DAD):** A novel diagnostic architecture is proposed that integrates MRI, PET, and cognitive scores while maintaining computational efficiency suitable for mobile and real-time clinical deployment.
- 2. Redundancy-Reduced Local and Global Feature Extraction:** The image feature extraction module employs spatial and channel redundancy reduction along with global frequency-domain filtering using Fast Fourier Transform (FFT), achieving high performance with fewer parameters.
- 3. Effective Multimodal Data Fusion Strategy:** A simple yet effective image fusion technique is applied by registering and combining segmented MRI and PET images to enrich structural and metabolic information while preserving hippocampal features critical for AD detection.
- 4. Use of Transfer Learning for Improved Generalization:** A parameter-sharing strategy based on transfer learning from the MobileViT network pretrained on ImageNet enhances model robustness and reduces training time on small medical datasets.
- 5. Loss Function Optimization for Imbalanced Data:** The study introduces a polynomial loss function tailored to mitigate class imbalance, leading to improved accuracy in both binary and multiclass classification tasks.

The subsequent sections of this manuscript are structured as follows: Section 2 delineates the proposed diagnostic methodology, encompassing data gathering, preprocessing procedures for imaging and cognitive scores, and the architecture of the Light-Mo-DAD network. Section 3 delineates the experimental configuration, assessment criteria, and comparison findings, emphasizing the diagnostic efficacy of the proposed model relative to established methodologies. Section 4 closes the study by summarizing the principal findings and delineating prospective future work focused on clinical implementation and the improvement of generalization via augmented datasets.

2. Diagnostic methods

2.1. Data collection

The data used comes from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database [20]. This database is a global research project designed to increase understanding of AD and related diseases through the use of neuroimaging, biomarkers, and clinical assessments. Diagnosing AD using a single neuroimaging modality makes it difficult to obtain informative features; using cognitive function scores obtained from patients in clinical examinations for diagnosis can yield rich multimodal features. In this study, T1-weighted MRI and FDG-PET neuroimaging data, as well as neurocognitive assessment scores, were used to diagnose AD. The imaging data came from 443 subjects, and the scoring data came from follow-up information at different stages of the same group of subjects. Not all brain disorders can be identified with brain imaging methods. For instance, certain mental conditions like anxiety or depression cannot always be detectable by brain imaging. Medical procedures that are needless or even hazardous may result from false positives from brain imaging. Neural activity may be dispersed in a more loosely defined network, however images of brains splotted with well-defined colorful regions are quite

Table 1
Subject Information in the ADNI Database.

Type	NM	NF	Age	SMMSE	SGDS	SCDR	SFAQ	SNPI
AD	64	41	75.76±24.25	21.48±8.52	1.93±6.12	0.93±0.40	16.68±4.23	4.59±4.86
CN	82	104	75.92±12.13	28.88±4.21	1.09±3.65	0.07±0.12	0.67±0.76	0.71±0.61
MCI	75	77	75.95±11.29	25.68±4.86	1.96±4.53	0.59±0.35	7.53±2.62	2.75±4.85

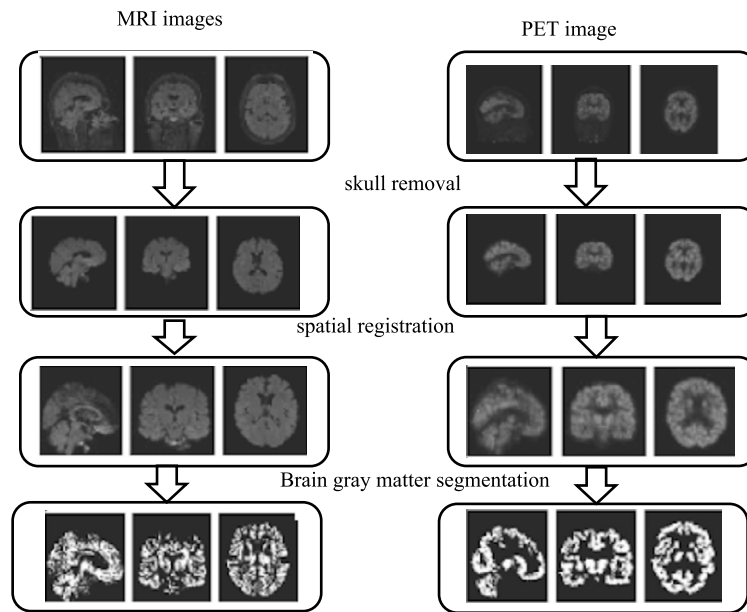


Fig. 1. Image preprocessing process.

deceptive since they imply well-defined processing modules (the module metaphor). Among AD patients, females were fewer than males, and younger patients were fewer than older ones; data was selected as evenly as possible. Demographic and clinical information of the subjects in the dataset is shown in Table 1. In the table, CN represents healthy controls, MCI represents patients with mild cognitive impairment, NM represents the number of males, NF represents the number of females, SMMSE represents the simplified mental state examination score, SGDS represents the global deterioration scale score, SCDR represents the clinical dementia rating scale score, SFAQ represents the functional activities questionnaire score, and SNPI represents the simplified neuropsychiatric inventory questionnaire score.

2.2. Data preprocessing

Medical imaging data and medical cognitive function assessment scores play an irreplaceable role in AD diagnosis. MRI images can provide structural information of the brain in AD patients, PET images can provide metabolic information of various brain tissues, and multimodal images that fuse these two types of images can simultaneously provide both structural and metabolic information of brain tissues. Patient cognitive state information obtained from cognitive function assessment scores is beneficial for AD diagnosis; preprocessing medical imaging data and medical cognitive function assessment scores can improve the accuracy of AD diagnosis.

2.2.1. Image data preprocessing

As shown in Fig. 1, image preprocessing mainly includes four parts: skull stripping, registration, segmentation, and fusion.

1. Using the BET tool in the medical image processing software FSL (FMRIB Software Library) [21], skull stripping was performed separately on patients' brain MRI and PET images. This

technique retains only intracranial tissue structures and removes unnecessary anatomical regions. The BET tool uses a fully automated algorithm to separate brain regions from surrounding non-brain tissues (such as skull, scalp, and meninges) in neuroimages [22]. It accurately estimates brain boundaries through methods such as threshold segmentation, surface fitting, and optimization techniques.

2. The FLIRT tool was used to register MRI and PET images to the MNI152 space. FLIRT aligns images with reference images by calculating rigid and affine transformations. These linear transformations can perform translation, rotation, and scaling while maintaining the morphological structure of the image, ensuring correspondence between the brain tissues of the two images.
3. The FAST tool in FSL was used to segment the gray matter regions of the brain in the images. FAST is a tool based on probabilistic models and Bayesian classifiers, which can estimate the spatial distribution of brain tissue types based on MRI images, including segmentation of gray matter, white matter, and cerebrospinal fluid, and respectively obtain images containing only gray matter and white matter.

Fusing the information of MRI and PET images can obtain a more comprehensive and accurate representation of brain structure and function. To this end, MRI and PET images were registered in the MNI152 standard space to ensure spatial alignment of the two types of images. After segmenting the two types of images respectively, gray matter, white matter, and cerebrospinal fluid images were obtained. The hippocampus is mainly composed of gray matter. In this study, the FLIRT component was used to register the gray matter image obtained from MRI image segmentation with the white matter image obtained from PET image segmentation to generate registered fused images. The gray matter image obtained from MRI segmentation retains the structural features of the

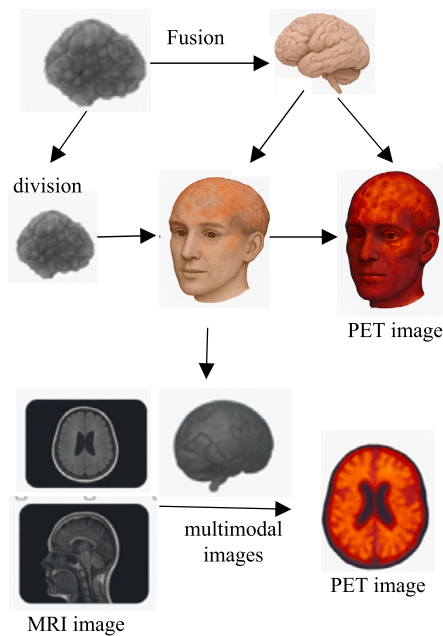


Fig. 2. Multimodal image fusion process.

hippocampus, and the white matter image obtained from PET segmentation retains the metabolic information in brain tissues. This fusion method is simple and practical, effectively preserving basic brain information while minimizing computational burden. By integrating rich semantic information and image details, this method improves the performance and accuracy of the model. The multimodal image fusion process is shown in Fig. 2.

MRI and PET images are usually stored in NIFTI format, which stores the three-dimensional data of MRI and PET [23]. Processing three-dimensional data requires high computational capability. In order to minimize the number of training parameters and computational complexity, two-dimensional data is used for training. Slicing in three directions – sagittal, coronal, and axial – can accurately detect and study AD-related changes [24], but not every slice can fully capture the feature information in the brain. In order to retain those parts containing key brain information, the obtained images need to be filtered in the study. As shown in Fig. 3, through the dimensionality reduction slicing process, multimodal image data containing key information from two types of images is obtained.

2.2.2. Text data preprocessing

The quality and accuracy of text data depend on the data source. The data source may be unreliable or incomplete, resulting in inaccurate analysis. Therefore, the obtained text data needs to be pre-processed. MMSE, GDS, CDR, FAQ, and NPI from patients in the ADNI database were selected as text data. Data cleaning and outlier processing operations were performed on the scoring information to remove missing data and abnormal data from the patient scoring data. The obtained data was normalized to eliminate the different measurement units and ranges between different scores, thereby obtaining the data used for training.

2.3. Diagnostic network

2.3.1. Overall network structure

Fig. 4 illustrates the structural diagram of the proposed network. This network aims to minimize the number of parameters while maintaining accuracy, establishing a basis for future expanded applications on mobile devices. Consequently, the configuration of the feature extraction module within the network

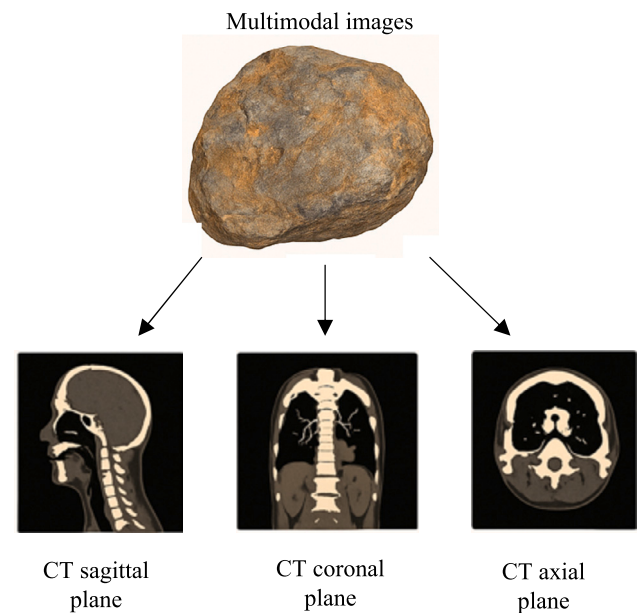


Fig. 3. Dimensionality Reduction Process of Multimodal Image Data.

primarily focuses on minimizing the parameter count. The inverted residual structure enhances network depth while diminishing complexity [25]. Subsequently, the image local feature extraction module and the global feature extraction module are employed to extract features from images. Using the Fast Fourier Transform (FFT) for global frequency-domain filtering and spatial and channel redundancy reduction, the image feature extraction module achieves great performance with fewer parameters. Using global feature extraction techniques, a single, representative set of features is extracted from an entire image or dataset. As opposed to local features, which concentrate on particular areas or patterns, these features capture the general qualities of the data. A separable convolutional network is employed to extract features from the scoring data. The extracted textual features are integrated with visual features to derive multimodal features. A linear classifier is ultimately employed for diagnostic classification.

2.3.2. Image feature extraction module

The feature extraction module combining image local features and global features is shown in Fig. 5. Features of the image are extracted from two dimensions spatial and channel to obtain local static features. The image features are extracted in the frequency domain and filtered by a global filter to obtain global features. The separately obtained image local features and global features are fused to obtain multimodal image features.

Local Feature Extraction In the traditional convolution process, the number of network parameters is often related to the size of the convolution kernel and the number of convolutions [26]. In order to obtain better local image features, researchers usually choose to use multiple convolutions or convolutions of different scales for feature extraction. Whether deepening the network depth or using convolutions of different scales will increase the number of network parameters. There is spatial and channel redundancy in the traditional convolution process. Therefore, compressing spatial and channel redundancy becomes the key factor in reducing the number of parameters. Spatial redundancy is compressed through separation and reconstruction, and channel redundancy is suppressed through separation and transformation fusion [27]. Through these two methods, a considerable performance improvement is achieved while reducing a large amount of computational

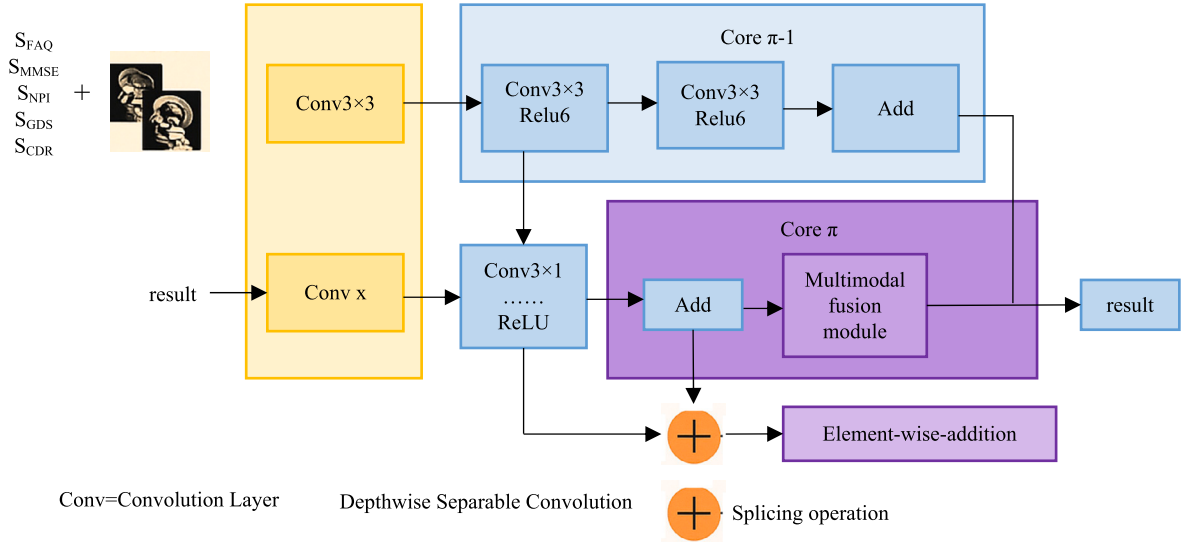


Fig. 4. Multimodal lightweight Alzheimer's disease auxiliary diagnosis network.

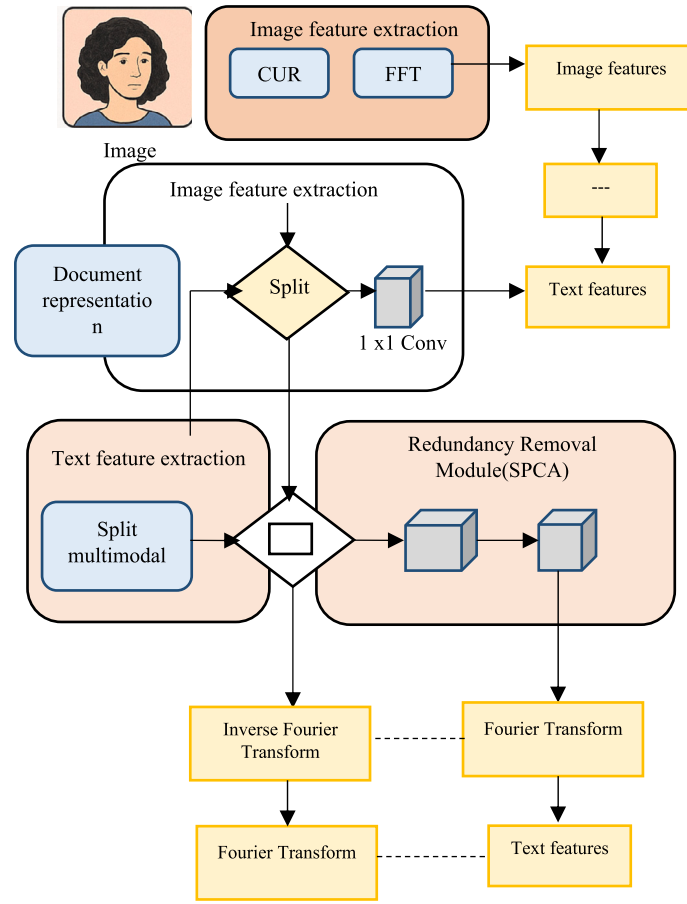


Fig. 5. Feature extraction module of the proposed network.

load. Spatial reconstruction first standardizes the original feature map using batch normalization:

$$X_{out} = \gamma \frac{X - \mu}{\sqrt{\sigma^2 + \varepsilon}} + \beta \quad (1)$$

Where X_{out} is the feature map after normalization, μ, σ are the mean and standard deviation of X , respectively, and ε is a small positive constant. γ and β are both trainable affine transforma-

tions. Separable convolutions are used in this module to extract features from cognitive score data. The amount of AD score data is enormous, and there are frequently a lot of parameters when using standard convolution for feature extraction. Feature extraction from three channels is not necessary for feature extraction of the score information. Numerous experimental results demonstrate that the optimum feature extraction performance is obtained when two separable convolutions are used throughout the text feature

extraction procedure. The trainable parameters $\gamma_1, \gamma_2, \gamma_3$ are used to measure the spatial pixel variance of each batch and channel. Rich spatial information reflects various changes in spatial pixels, leading to larger values of γ . The obtained feature mapping parameters are normalized to obtain the normalized feature weight parameters Z_γ , which represent the importance of different feature maps:

$$Z_\gamma = \frac{\gamma}{1_c^U \gamma}. \quad (2)$$

In the formula, $\mathbf{c}$ represents the channel size. The standardized original data is concatenated with the feature weights obtained through feature mapping and normalization, resulting in feature weights that contain diverse feature information.

$$X_C = X \otimes Z_\gamma \quad (3)$$

The feature weights are then processed with Sigmoid activation and gated using a threshold. Weights above the threshold are set to 1, resulting in Z_1 ; weights below the threshold are set to 0.5, resulting in Z_2 . There are no cases where the weights are exactly equal to the threshold.

$$Z = \text{Gate}(\text{Sigmoid}(X_C)) \quad (4)$$

Z_1 and Z_2 are each multiplied with the original image to obtain feature maps X_1^Z and X_2^Z , which contain rich and limited information, respectively. To reduce spatial redundancy through cross-reconstruction, X_1^Z and X_2^Z are each divided into two parts based on the number of channels, resulting in $X_{11}^Z, X_{12}^Z, X_{21}^Z, X_{22}^Z$. X_{11}^Z is added to X_{21}^Z , and X_{12}^Z is added to X_{22}^Z yielding X^{1Z} and X^{2Z} respectively. X^{1Z} and X^{2Z} are then concatenated, combining the features rich in information with those containing less information, to obtain the spatially constrained feature map X^Z .

$$X^Z = \text{Concat}(X^{1Z}, X^{2Z}) \quad (5)$$

Channel reconstruction is performed on X^Z by splitting the channels, where α is the splitting ratio $\alpha \in (0, 1)$ and C is the number of input channels. This results in feature maps with αC channels and $(1 - \alpha)C$ channels. A 1×1 convolution is applied to compress the feature maps in order to improve computational efficiency. After the splitting and compression operations, the original feature map X^Z is divided into two parts: X_{up} and $X_{\text{low}2}$.

$$X^Z = \text{Add}(X_{\text{up}}, X_{\text{low}}) \quad (6)$$

The feature map X_{up} with αC channels is processed using a combination of group convolution and pointwise convolution to extract features, resulting in $X_{\text{up}}^{\text{GC}}$ and $X_{\text{up}}^{\text{PC}}$ respectively. Group convolution reduces the number of parameters and computational load, while pointwise convolution enhances the information flow between channel groups. The features extracted from these two types of convolutions are then added together to obtain the feature map Y_1 .

$$Y_1 = X_{\text{up}}^{\text{GC}} \otimes X_{\text{up}}^{\text{PC}} \quad (7)$$

The feature map $X_{\text{low}2}$ with $(1 - \alpha)C$ channels is concatenated with the feature map $X_{\text{low}}^{\text{PC}}$ which has been processed by pointwise convolution, to obtain the feature map Y_2 .

$$Y_2 = \text{Concat}(X_{\text{low}2}^{\text{PC}}, X_{\text{low}}^{\text{PC}}) \quad (8)$$

Global channel descriptors T_1 and T_2 are obtained by applying pooling operations to the feature maps Y_1 and Y_2 , which have been processed in the upper and lower branches, respectively.

$$T_m = \frac{1}{K \times Z} \sum_{i=1}^K \sum_{j=1}^Z Y_C(i, j); \quad m = 1, 2 \quad (9)$$

The feature importance vectors β_1 and β_2 are generated using T_1 and T_2 .

$$\beta_1 = \frac{\exp S_1}{\exp S_1 + \exp S_2}, \quad \beta_2 = \frac{\exp S_2}{\exp S_1 + \exp S_2}, \quad \beta_1 + \beta_2 = 1 \quad (10)$$

β_1 and β_2 are combined with Y_1 and Y_2 to obtain the final channel-constrained feature Y .

$$Y = \beta_1 Y_1 + \beta_2 Y_2 \quad (11)$$

These two methods spatial reconstruction and channel reconstruction – maximize the reduction of channel and spatial redundancy generated during feature extraction, which also reduces the number of parameters required for local feature extraction.

(1) Global Feature Extraction

In the field of deep learning, capturing the global features of data typically considers the Transformer architecture. The self-attention mechanism and multilayer network structure of the Transformer increase the parameters of the entire network [28]. By transforming the features of image data into the frequency domain for extraction, the number of parameters can be reduced [29]. The Fast Fourier Transform (FFT) is used to convert the feature $X \in \mathbf{R}^{K \times Z \times C}$ of the image data into the frequency domain, obtaining the original feature:

$$Y(\mu, w) = \sum_{h=0}^{K-1} \sum_{z=0}^{Z-1} X(h, z) \exp(-2\pi(\mu h, wz)) \quad (12)$$

A Multilayer Perceptron (MLP) is used for feature extraction, yielding the extracted feature Z :

$$Z = Z_n (Z_{n-1} (\cdots (Z_1 X + b_1) \cdots) + b_n). \quad (13)$$

A 1×1 convolution (equivalent to a linear transformation) is used to obtain a mask tensor $\mathbf{M}$, which contains the global features of the original image information from the original image. The features learned from the frequency domain are globally filtered through element-wise multiplication:

$$M(\mu, w) = ZX(h, z) + b \quad (14)$$

$$N = M \odot Z \quad (15)$$

The Inverse Fast Fourier Transform (IFFT) is used to convert the filtered feature $\mathbf{N}$ back to the original domain, obtaining the final feature $\hat{\mathbf{X}}$:

$$\hat{X} = F^{-1}[N]. \quad (16)$$

The Fast Fourier Transform enables this feature extraction method to outperform the Vision Transformer both in terms of parameter count and computational speed. Transformer-based architectures are a type of deep learning technique that does not rely on recurrent or convolutional layers; instead, it uses the multi-head self-attention mechanism to analyze sequential data, such text, or images. This enables parallel processing, which improves performance and speeds up training, especially for tasks with long-range dependencies. The Transformer's multilayer network topology and self-attention mechanism raise the network's overall parameters.

During the feature extraction process, since the mask tensor is obtained via 1×1 convolution (which can be regarded as a global convolution), the filtered features all contain contextual information of the image, achieving the same global feature extraction capability as the Vision Transformer [30].

2.3.3. Text feature extraction module

This is a module that extracts cognitive score information features using separable convolutions. The quantity of AD score information is massive, and if traditional convolution is used for feature extraction, it often results in a large number of parameters. Feature extraction of the score information does not involve feature extraction from three channels. Therefore, separable convolution ([31–34]) is used to extract text information features in order to reduce network parameters. Depthwise separable convolution consists of two parts:

1. Depthwise Convolution

$$G_{i,j,c} = \sum_{m=1}^{K_h} \sum_{n=1}^{K_z} A_{i+m,j+n,c} K_{m,n,c} \quad (17)$$

In the equation: $G_{i,j,c}$ is the output tensor of the depthwise convolution, $A_{i+m,j+n,c}$ is the input text data tensor, $K_{m,n,c}$ is the position of the convolution kernel in input channel c , K_h and K_z are the height and width of the convolution kernel, respectively.

2. Pointwise Convolution

$$Y_{i,j,k} = \sum_{c=1}^C G_{i,j,c} Q_{1,1,c,k} \quad (18)$$

In the equation: $Y_{i,j,k}$ is the final output of the depthwise separable convolution, k is the number of output channels, and $Q_{1,1,c,k}$ is the weight of the pointwise convolution kernel. During the process of text feature extraction, a large number of experimental results show that using two separable convolutions yields the best feature extraction performance.

2.4. Loss function

2.4.1. Cross-entropy loss function

For binary classification problems, assume the model's output is a probability value, mapped to the range $[0, 1]$ using the Sigmoid function. If the true label is 1, the cross-entropy loss is expressed as

$$L_{BCE}(y, \hat{y}) = -(y \lg \hat{y} + (1 - y) \lg(1 - \hat{y})) \quad (19)$$

where: y is the true label, $\hat{y}$ is the predicted label. For multi-class classification problems, assume the model has K categories, and the output is normalized into a probability distribution by the Softmax function. If the sample belongs to the i class, the cross-entropy loss is expressed as

$$L_{BCE} = (y, \hat{y}) = - \sum_{i=1}^K y_i \lg \hat{y}_i, \quad (20)$$

where: y is the i element of the true label, $\hat{y}$ is the i element of the model's predicted probability distribution.

2.4.2. Focal loss function

This function is used for dense object detection, and is expressed as

$$FL(q_t) = -\alpha_t (1 - q_t)^\gamma \lg q_t, \quad (21)$$

where: q_t is the model's predicted probability that the sample belongs to the positive class; α_t is the balancing factor, used to adjust the importance of positive and negative samples, and is set based on the actual situation; γ is the adjustable focusing parameter, used to adjust the focus on hard-to-classify samples. The focal loss function adjusts the weights of the loss function to pay more attention to hard-to-classify samples. When a sample belongs to an easily classified category, q_t is close to 1, so $(1 - q_t)$ is close to 0, thereby reducing the weight of the sample; for hard-to-classify samples, since the weight of the loss function is enhanced, these samples receive more attention. In the case of class imbalance, the focal loss function performs well in some object detection tasks and some classification tasks.

2.4.3. Polynomial loss function

The polynomial loss function offers a polynomial expansion framework for comprehending and enhancing the widely utilized cross-entropy loss and focal loss functions. The author [35] examined the fundamental nature of common cross-entropy loss and focal loss functions via Taylor series expansion and derivation, introducing the concepts of vertical regulation loss and horizontal regulation loss. They examined the mechanism of data imbalance in loss functions and presented straightforward yet effective solutions to address the issue, attaining results that exceed those of the cross-entropy and focal loss functions. The expression of the polynomial loss function is

$$L_q = \sum_{j=1}^{\infty} \alpha_j (1 - Q_t)^j \quad (22)$$

Using the complete polynomial loss function increases hyper parameters, as well as parameter count and computational complexity. To reduce the number of parameters, the first polynomial coefficient is retained as the loss function:

$$L_{p1} = L_{BCE} + \varepsilon_1 (1 - Q_t) \quad (23)$$

$$L_{p1} = (\varepsilon_1 + 1) (1 - Q_t) + \frac{1}{2} (1 - Q_t)^2 + \dots + \frac{1}{N} (1 - Q_t)^N + \frac{1}{1+N} (1 - Q_t)^{N+1} + \dots = -\lg(Q_t) + \varepsilon_1 (1 - Q_t) \quad (24)$$

This loss function has maximum gain, requires minimal code modifications and hyperparameter adjustments, and meets the requirements of a lightweight network for this study. It will be used as the network loss function in this study.

2.5. Transfer learning

The complexity of medical data collection makes it difficult to obtain large medical datasets [36]. In the field of deep learning, small datasets can lead to overfitting, model instability, limited feature representation capability, decreased generalization ability, and inability to handle complex tasks. Small datasets in deep learning can result in overfitting, unstable models, poor feature representation, poor generalization, and an inability to manage challenging tasks. Transfer learning makes use of previously acquired knowledge to enhance the process of learning new activities. By sharing commonalities between the source and target domains, the fundamental concept is to apply the model parameters, feature representations, or knowledge acquired in the source domain to the target job. With a small dataset, this study produced comparatively good diagnostic results. This study achieved relatively ideal diagnostic results using a limited dataset. As shown in Fig. 6, the parameter-based method in transfer learning is adopted. A model is built

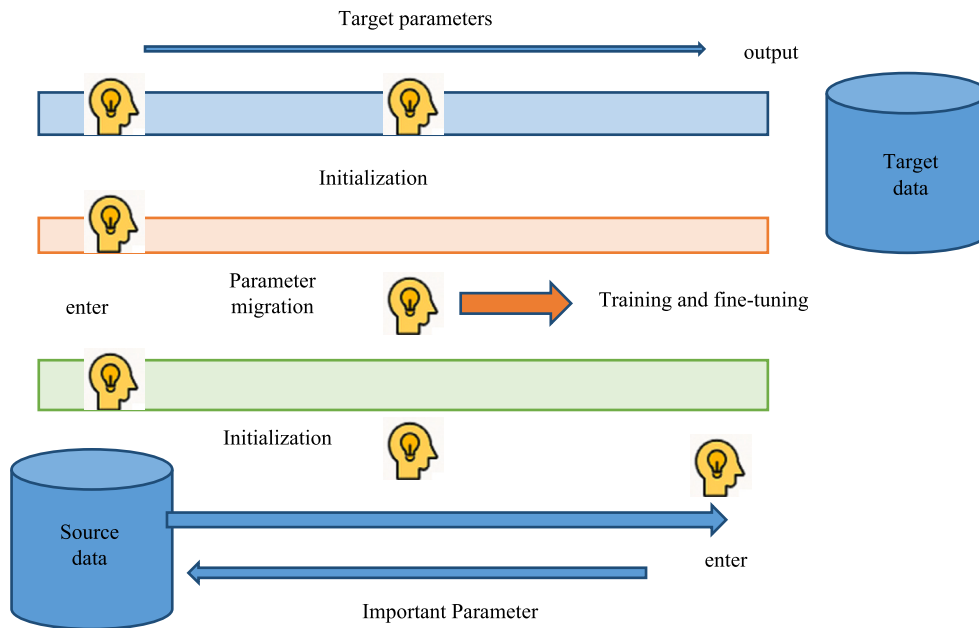


Fig. 6. Schematic diagram of transfer learning.

using the source data, and then the learned model parameters are shared with the target domain as a pre-obtained model, which is used for training and fine-tuning of parameters. The source data in this study comes from the ImageNet 1k dataset, which contains approximately 10 million images covering 1,000 categories, including various domains such as animals, plants, objects, and scenes. The model parameters were obtained by training on the same MobileViT [37] network structure proposed in this study using this dataset. The model parameters were acquired by training on the same MobileViT network structure presented in this study utilizing this dataset. Transfer learning can significantly cut down on the amount of time needed to train on different datasets and enhance the network's capacity for generalization. To categorize and identify topics, various lightweight networks were used, and the binary classification results were shown. Although MobileViT's classification accuracy is higher than that of the suggested network, its network computation load (FLOPS), number of parameters (NP), and memory usage (MAC) are significantly higher.

3. Experimental detail and result analysis

The proposed network was implemented using Python 3.7 and the PyTorch 1.7 framework. Model training was conducted on a workstation equipped with an Intel Core i9-10900X CPU and an NVIDIA RTX 3090 GPU. Transfer learning and extensive fine-tuning were applied to optimize performance, with the learning rate set to 0.001, a batch size of 32, and 150 training epochs. To reduce computational complexity and processing time, input images were cropped and sampled to a resolution of 128×128 pixels. This ensured the preservation of critical brain regions while enabling faster and more efficient training. Obtaining huge medical datasets is challenging due to the intricacy of medical data collection. Small datasets in deep learning can result in overfitting, unstable models, poor feature representation, poor generalization, and an inability to manage challenging tasks. Transfer learning makes use of previously acquired knowledge to enhance the process of learning new activities. By sharing commonalities between the source and target domains, the fundamental concept is to apply the model parameters, feature representations, or knowledge acquired in the source domain to the target job. The network's diagnostic performance was evaluated using both binary classification (AD vs.

Table 2

Classification results of different data models.

Model	ACC3	ACC2	SEN2	SPE2
MRI	0.73108	0.86436	0.8575	0.83104
PET	0.6713	0.90944	0.92708	0.90846
MRI+PET	0.80556	0.92414	0.93884	0.91532
Text	0.8624	0.92904	0.93492	0.92512
This study	0.89376	0.9604	0.9653	0.9555

CN) and three-class classification (AD, CN, and MCI) tasks. Accuracy (ACC), sensitivity (SEN), and specificity (SPE) were employed as performance metrics. To assess the impact of multimodal data integration, the model was tested on various combinations of input data—MRI, PET, combined MRI+PET, cognitive scores, and a full fusion of all modalities. As reported in Table 2, single-modality inputs (MRI or PET) achieved relatively lower accuracy compared to the fused multimodal inputs, validating the effectiveness of the proposed fusion strategy. Incorporating cognitive scores alongside imaging data led to a 0.032 improvement in three-class accuracy compared to using scores alone, and improvements of 0.037 and 0.090 over using MRI or PET alone, respectively. These results highlight the diagnostic value of integrating cognitive assessments with neuroimaging, reinforcing the robustness and reliability of the proposed multimodal approach for Alzheimer's disease diagnosis.

Different lightweight networks were selected to classify and identify subjects, and the results of binary classification are shown in Table 3. As can be seen from the table, MobileViT has better classification accuracy, but the network computation load (FLOPS), number of parameters (NP), and memory consumption (MAC) are much higher than those of the proposed network. The MAC of Mobileswin is similar to that of the proposed network, but its other indicators are inferior to the proposed network. Using the proposed network with only text features, both NP and MAC are much lower than when using image features. The reason is that the image data is much larger than the scoring data, and using a deeper network for image feature extraction yields better results. Clinically, patients show increasing memory loss, impaired cognitive function, and confusion, often leading to complete dependence. Despite significant advancements in neuroscience, the complexities of the disease's mechanism and symptom overlap with other cognitive illnesses make it challenging to diagnose Alzheimer's dis-

Table 3
Performance parameters of different classification networks.

Network	ACC	SEN	SPE	FLOPS\10 ⁸	NP\10 ⁵	MAC\10 ⁸
MobileViT [37]	0.92806	0.94864	0.92806	3.234	9.702	9.016
Mobileswin [38]	0.90454	0.9261	0.93296	2.058	4.802	7.252
Image features [39]	0.92414	0.93884	0.91532	1.274	3.234	7.448
Text features [40]	0.92904	0.93492	0.92512	3.136	2.646	1.568
Light-Mo-DAD	0.9604	0.9653	0.9555	1.274	3.234	7.448

Table 4
Classification results of different loss functions.

Loss function	ACC ₃	ACC ₂	SEN ₂	SPE ₂
Cross entropy	0.83888	0.94178	0.94766	0.93688
Focus	0.87318	0.95256	0.96824	0.94864
Polynomial	0.89376	0.9604	0.9653	0.9555

ease accurately and promptly. The FLOPS and NP of the proposed network are smaller than those of MobileViT and Mobileswin [41], and the incorporation of multimodal data ensures the diagnostic performance of the network [42–44].

Lightweight networks often suffer from overfitting during training, which is caused by high network complexity and imbalanced data distribution. Although this study paid attention to balanced distribution during the data collection phase, the variability of the data still makes it possible for imbalance to arise after preprocessing. To explore the effect of different loss functions on network training, different loss functions were selected for training, and the training results are shown in Table 4. As can be seen from the table, the polynomial loss function improved the binary classification accuracy by 0.019 and 0.008 compared to the cross-entropy loss function and focal loss function, respectively, and improved the three-class classification accuracy by 0.056 and 0.021, respectively. These results verify that in training tasks with imbalanced data, the polynomial loss function performs better.

4. Conclusion

This study presents Light-Mo-DAD, a lightweight multimodal neural network that integrates neuroimaging modalities (MRI and PET) with neuropsychological assessment scores to support the early and accurate diagnosis of Alzheimer's Disease (AD). Rooted in neuroscience, the model addresses the complex neuropathological underpinnings of AD, including hippocampal atrophy, cortical thinning, and metabolic dysfunction—hallmarks observable through T1-weighted MRI and FDG-PET scans. By employing redundancy-reduced convolutions and a frequency-domain global feature extraction mechanism, Light-Mo-DAD effectively captures both local structural abnormalities and global functional disruptions in the brain. Moreover, the fusion of imaging features with cognitive scores reflects the multifactorial nature of AD pathology, enhancing diagnostic reliability. The model's use of depthwise separable convolutions for processing cognitive data aligns with the need to handle diverse neurobehavioral metrics efficiently, such as those captured by MMSE, CDR, and FAQ scores. The incorporation of transfer learning from ImageNet-trained MobileViT ensures robust feature generalization, which is essential given the limited size and variability of clinical datasets. Experimental results on the ADNI dataset demonstrate significant improvements in classification performance, particularly in distinguishing among cognitively normal individuals, patients with mild cognitive impairment, and those with AD. By integrating structural, metabolic, and cognitive dimensions of AD, Light-Mo-DAD aligns with current neuroscientific understanding of disease progression and provides a practical, low-resource solution for early-stage diagnosis. Future research will concentrate on extending clinical partnerships to incorporate real-world cognitive tests and longitudinal neuroimaging

data. Furthermore, by incorporating the model into mobile clinical tools, neurologists may make educated judgments based on a multimodal, pathophysiologically grounded framework, facilitating point-of-care screening. Given the small size and unpredictability of clinical datasets, robust feature generalization is ensured by incorporating transfer learning from ImageNet-trained MobileViT.

CRedit authorship contribution statement

Bhoomi Gupta: Conceptualization, Data curation, Methodology, Resources, Software, Writing – original draft. **Ganesh Kanna Jegannathan:** Conceptualization, Data curation, Methodology, Resources, Software, Writing – original draft. **Mohammad Shabbir Alam:** Conceptualization, Data curation, Methodology, Resources, Software, Writing – original draft. **Kottala Sri Yogi:** Conceptualization, Data curation, Methodology, Resources, Software, Writing – original draft. **Janjhyam Venkata Naga Ramesh:** Formal analysis, Investigation, Validation, Visualization, Writing – review & editing. **Vemula Jasmine Sowmya:** Formal analysis, Investigation, Validation, Visualization, Writing – review & editing. **Isa Bayhan:** Formal analysis, Investigation, Validation, Visualization, Writing – review & editing.

Informed consent and patient details

The authors declare that the work described does not involve patients or volunteers.

Human and animal rights

The authors declare that the work described has not involved experimentation on humans or animals.

Funding

This work did not receive any grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial or personal relationships that could be viewed as influencing the work reported in this paper.

Data availability

Data shall be available on request.

References

- [1] Asrar G. Alharthi, Salha M. Alzahrani, Do it the transformer way: a comprehensive review of brain and vision transformers for autism spectrum disorder diagnosis and classification, *Comput. Biol. Med.* (ISSN 0010-4825) 167 (2023) 107667, <https://doi.org/10.1016/j.cmpbiomed.2023.107667>.
- [2] Sajid Nazir, Diane M. Dickson, Muhammad Usman Akram, Survey of explainable artificial intelligence techniques for biomedical imaging with deep neural networks, *Comput. Biol. Med.* (ISSN 0010-4825) 156 (2023) 106668, <https://doi.org/10.1016/j.cmpbiomed.2023.106668>.

- [3] A. Kumar, S. Pradeep, K. Arora, G. Sreeram, A. Pankajam, T. Patil, A. Sahu, Feature fusion based deep learning model for Alzheimer's neurological disorder classification, *Neurosci. Inf.* 5 (2) (2025) 100196, <https://doi.org/10.1016/j.neuri.2025.100196>.
- [4] A. Sharma, S. Kaur, N. Memon, A. Jainul Fathima, S. Ray, M.W. Bhatt, Alzheimer's patients detection using support vector machine (SVM) with quantitative analysis, *Neurosci. Inf.* 1 (3) (2021) 100012, <https://doi.org/10.1016/j.neuri.2021.100012>.
- [5] M. Orouskhani, C. Zhu, S. Rostamian, F. Shomal Zadeh, M. Shafiei, Y. Orouskhani, Alzheimer's disease detection from structural MRI using conditional deep triplet network, *Neurosci. Inf.* 2 (4) (2022) 100066, <https://doi.org/10.1016/j.neuri.2022.100066>.
- [6] S. Kazemi-Harikandei, P. Shobeiri, M. Salmani Jelodar, S.M. Tavangar, Effective connectivity in individuals with Alzheimer's disease and mild cognitive impairment: a systematic review, *Neurosci. Inf.* 2 (4) (2022) 100104, <https://doi.org/10.1016/j.neuri.2022.100104>.
- [7] Bhanu Priya, Pranav Gupta, Shantanu Singh, Chapter 19 - Harnessing the potentials of machine learning models in Alzheimer's disease prediction and detection, in: Hari M. Srivastava, Geeta Arora, Firdous A. Shah (Eds.), *Advances in Computational Methods and Modeling for Science and Engineering*, Morgan Kaufmann, ISBN 9780443300127, 2025, pp. 259–267.
- [8] Farhatullah, Xin Chen, Deze Zeng, Atif Mehmood, Rizwan Khan, Farah Shahid, Mostafa M. Ibrahim, 3-Way hybrid analysis using clinical and magnetic resonance imaging for early diagnosis of Alzheimer's disease, *Brain Res.* (ISSN 0006-8993) 1840 (2024) 149021, <https://doi.org/10.1016/j.brainres.2024.149021>.
- [9] Li Xiaoxue, Zhang Xiaofan, Yi Xin, Liu Dan, Wang He, Zhang Bowen, Zhang Bohan, Zhao Di, Wang Liqun, Review of medical data analysis based on spiking neural networks, *Proc. Comput. Sci.* (ISSN 1877-0509) 221 (2023) 1527–1538, <https://doi.org/10.1016/j.procs.2023.08.138>.
- [10] Karim Aderghal, Karim Afdel, Jenny Benois-Pineau, Gwénaëlle Catheline, Improving Alzheimer's stage categorization with Convolutional Neural Network using transfer learning and different magnetic resonance imaging modalities, *Heliyon* (ISSN 2405-8440) 6 (12) (2020) e05652, <https://doi.org/10.1016/j.heliyon.2020.e05652>.
- [11] Nasir Rahim, Naveed Ahmad, Waseem Ullah, Jatin Bedi, Younhyun Jung, Early progression detection from MCI to AD using multi-view MRI for enhanced assisted living, *Image Vis. Comput.* (ISSN 0262-8856) 157 (2025) 105491, <https://doi.org/10.1016/j.imavis.2025.105491>.
- [12] Md. Ashif Mahmud Joy, Shamima Nasrin, Ayesha Siddiqua, Dewan Md. Farid, ViTAD: leveraging modified vision transformer for Alzheimer's disease multi-stage classification from brain MRI scans, *Brain Res.* (ISSN 0006-8993) 1847 (2025) 149302, <https://doi.org/10.1016/j.brainres.2024.149302>.
- [13] Zhaomin Yao, Weiming Xie, Jiaming Chen, Ying Zhan, Xiaodan Wu, Yingxin Dai, Yusong Pei, Zhiguo Wang, Guoxu Zhang, IT: an interpretable transformer model for Alzheimer's disease prediction based on PET/MR images, *NeuroImage* (ISSN 1053-8119) 311 (2025) 121210, <https://doi.org/10.1016/j.neuroimage.2025.121210>.
- [14] S. Putha, S.R. Gayam, B.P. Kasaraneni, K.K. Kondapaka, S.K. Nallamala, P. Thuniki, Neuroscience-informed nomogram model for early prediction of cognitive impairment in Parkinson's disease, *Neurosci. Inf.* 5 (2) (2025) 100189.
- [15] I.H. Elshekhdri, M.B. Amien, A. Fragoon, Automated classification of epileptic seizures using modified one-dimensional convolution neural network based on empirical mode decomposition with high accuracy, *Neurosci. Inf.* 5 (2) (2025) 100188.
- [16] H.I. Hidayatullah, M.T. Saifullah, M.T. Ghazali, A. Aziz, Exploring community pharmacist's psychological intentions to adopt generative artificial intelligence (GenAI) chatbots for patient information, education, and counseling, *Neurosci. Inf.* 20213 (2025).
- [17] S.K. Dey, K.M.M. Uddin, A. Howlader, M.M. Rahman, H.M.H. Babu, N. Biswas, U.R. Siddiqi, B. Mazumder, Analyzing infant cry to detect birth asphyxia using a hybrid CNN and feature extraction approach, *Neurosci. Inf.* 5 (2) (2025) 100193.
- [18] Md Assaduzzaman, Monoronjon Dutta, Arpa Saha, Showmick Guha Paul, ALSA-3: customized CNN model through ablation study for Alzheimer's disease classification, *Inform. Med. Unlocked* (ISSN 2352-9148) 50 (2024) 101584, <https://doi.org/10.1016/j.imu.2024.101584>.
- [19] Jiayi Zhu, Ying Tan, Rude Lin, Jiaqing Miao, Xuwei Fan, Yafei Zhu, Ping Liang, Jinnan Gong, Hui He, Efficient self-attention mechanism and structural distilling model for Alzheimer's disease diagnosis, *Comput. Biol. Med.* (ISSN 0010-4825) 147 (2022) 105737, <https://doi.org/10.1016/j.combiomed.2022.105737>.
- [20] Rahma Kadri, Bassem Bouaziz, Mohamed Tmar, Faiez Gargouri, Innovative multi-modal approaches to Alzheimer's disease detection: transformer hybrid model and adaptive MLP-Mixer, *Pattern Recognit. Lett.* (ISSN 0167-8655) 190 (2025) 15–21, <https://doi.org/10.1016/j.patrec.2025.01.029>.
- [21] Nasir Rahim, Tamer Abuhmed, Seyedali Mirjalili, Shaker El-Sappagh, Khan Muhammad, Time-series visual explainability for Alzheimer's disease progression detection for smart healthcare, *Alex. Eng. J.* (ISSN 1110-0168) 82 (2023) 484–502, <https://doi.org/10.1016/j.aej.2023.09.050>.
- [22] Nasir Rahim, Shaker El-Sappagh, Sajid Ali, Khan Muhammad, Javier Del Ser, Tamer Abuhmed, Prediction of Alzheimer's progression based on multimodal Deep-Learning-based fusion and visual Explainability of time-series data, *Inf. Fusion* (ISSN 1566-2535) 92 (2023) 363–388, <https://doi.org/10.1016/j.inffus.2022.11.028>.
- [23] Khosro Rezaee, Min Zhu, Diagnose Alzheimer's disease and mild cognitive impairment using deep CascadeNet and handcrafted features from EEG signals, *Biomed. Signal Process. Control* (ISSN 1746-8094) 99 (2025) 106895, <https://doi.org/10.1016/j.bspc.2024.106895>.
- [24] Gehan Fatima, Akm Ashiquzzaman, Sang Seong Kim, Young Ro Kim, Hyuk-Sang Kwon, Euiheon Chung, Vascular and glymphatic dysfunction as drivers of cognitive impairment in Alzheimer's disease: insights from computational approaches, *Neurobiol. Dis.* (ISSN 0969-9961) 208 (2025) 106877, <https://doi.org/10.1016/j.nbd.2025.106877>.
- [25] Shashank Venkat, Tanmay Ghodeswar, Parth Chavan, Senthil Kumar Narayanasamy, Kathiravan Srinivasan, MRI-based automated diagnosis of Alzheimer's disease using Alzh-Net deep learning model, *Biomed. Signal Process. Control* (ISSN 1746-8094) 102 (2025) 107367, <https://doi.org/10.1016/j.bspc.2024.107367>.
- [26] Mahak Lashkary, Kanika Bansal, Samyak Jain, Praveen Kumar Shukla, Hitesh Tekchandani, Vijaypal Singh Dhaka, Detection of Alzheimer's disease progression from structural MRIs using ConvMixer based deep learning model, *Proc. Comput. Sci.* (ISSN 1877-0509) 258 (2025) 420–429, <https://doi.org/10.1016/j.procs.2025.04.278>.
- [27] M. Khojaste-Sarakhsi, Seyedhamidreza Shahabi Haghighi, S.M.T. Fatemi Ghomi, Elena Marchiori, Deep learning for Alzheimer's disease diagnosis: a survey, *Artif. Intell. Med.* (ISSN 0933-3657) 130 (2022) 102332, <https://doi.org/10.1016/j.artmed.2022.102332>.
- [28] Md. Sakib Bin Alam, Aiman Lameesa, Senzuti Sharmin, Shaila Afrin, Shams Forruque Ahmed, Mohammad Reza Nikoo, Amir H. Gandomi, Role of deep learning in cognitive healthcare: wearable signal analysis, algorithms, benefits, and challenges, *Digit. Commun. Netw.* (ISSN 2352-8648) (2025), <https://doi.org/10.1016/j.dcan.2025.04.001>.
- [29] Zhiwei Qin, Zhao Liu, Qihao Guo, Ping Zhu, 3D convolutional neural networks with hybrid attention mechanism for early diagnosis of Alzheimer's disease, *Biomed. Signal Process. Control* (ISSN 1746-8094) 77 (2022) 103828, <https://doi.org/10.1016/j.bspc.2022.103828>.
- [30] Nasir Rahim, Shaker El-Sappagh, Haytham Rizk, Omar Amin El-serafy, Tamer Abuhmed, Information fusion-based Bayesian optimized heterogeneous deep ensemble model based on longitudinal neuroimaging data, *Appl. Soft Comput.* (ISSN 1568-4946) 162 (2024) 111749, <https://doi.org/10.1016/j.asoc.2024.111749>.
- [31] Jin Gu, Xinbei Zha, Jiaming Zhang, Xiaole Zhao, Assisted diagnosis of neuropsychiatric disorders based on functional connectivity: a survey on application and performance evaluation of graph neural network, *Expert Syst. Appl.* (ISSN 0957-4174) 265 (2025) 125922, <https://doi.org/10.1016/j.eswa.2024.125922>.
- [32] Siwei Xie, Li Xiao, Haitao Huang, Dayang Chen, Haiman Guo, Amar Jain, Attention-augmented stockwell transform and convolutional neural network framework for electroencephalogram-based multi-class classification of Frontotemporal Dementia, *Inf. Fusion* (ISSN 1566-2535) 123 (2025) 103267, <https://doi.org/10.1016/j.inffus.2025.103267>.
- [33] Ahmed Elazab, Changmiao Wang, Mohammed Abdelaziz, Jian Zhang, Jason Gu, Juan M. Gorri, Yudong Zhang, Chunqi Chang, Alzheimer's disease diagnosis from single and multimodal data using machine and deep learning models: achievements and future directions, *Expert Syst. Appl.* (ISSN 0957-4174) 255, Part C (2024) 124780, <https://doi.org/10.1016/j.eswa.2024.124780>.
- [34] Qiang Zheng, Pengzhi Nan, Yongchao Cui, Lin Li, ConnectomeAE: multimodal brain connectome-based dual-branch autoencoder and its application in the diagnosis of brain diseases, *Comput. Methods Programs Biomed.* (ISSN 0169-2607) 267 (2025) 108801, <https://doi.org/10.1016/j.cmpb.2025.108801>.
- [35] Rong Zhang, Jinhua Sheng, Qiao Zhang, Junmei Wang, Binbing Wang, A review of multimodal fusion-based deep learning for Alzheimer's disease, *Neuroscience* (ISSN 0306-4522) 576 (2025) 80–95, <https://doi.org/10.1016/j.neuroscience.2025.04.035>.
- [36] Jalal Safari Bazargani, Nasir Rahim, Abolghasem Sadeghi-Niaraki, Tamer Abuhmed, Houbing Song, Soo-Mi Choi, Alzheimer's disease diagnosis in the metaverse, *Comput. Methods Programs Biomed.* (ISSN 0169-2607) 255 (2024) 108348, <https://doi.org/10.1016/j.cmpb.2024.108348>.
- [37] K.H. Aqil, Prashanth Dumpuri, Keerthi Ram, Mohanasankar Sivaprakasam, Predictive modeling of Alzheimer's disease progression: integrating temporal clinical factors and outcomes in time series forecasting, *Intell.-Based Med.* (ISSN 2666-5212) 10 (2024) 100159, <https://doi.org/10.1016/j.ibmed.2024.100159>.
- [38] Zhengning Wang, Jiaxin Liu, Fang Chen, Ke Wang, Yuhang Xu, Fengmei Lu, Yanling Li, Jingjing Gao, Investigating the relationship between brain age and Alzheimer's disease: a deep learning approach with multimodal MRI, *Biomed. Signal Process. Control* (ISSN 1746-8094) 109 (2025) 107926, <https://doi.org/10.1016/j.bspc.2025.107926>.
- [39] Yibo Huang, Jie Liu, Zhiyong Li, Qiuyu Zhang, MFIFN: a multimodal feature interaction fusion network-based model for Alzheimer's disease classification, *Biomed. Signal Process. Control* (ISSN 1746-8094) 107 (2025) 107857, <https://doi.org/10.1016/j.bspc.2025.107857>.

- [40] Hesheng Huang, Witold Pedrycz, Kaoru Hirota, Fei Yan, A multiview-slice feature fusion network for early diagnosis of Alzheimer's disease with structural MRI images, *Inf. Fusion* (ISSN 1566-2535) 119 (2025) 103010, <https://doi.org/10.1016/j.inffus.2025.103010>.
- [41] Uttam Khatri, Goo-Rak Kwon, Diagnosis of Alzheimer's disease via optimized lightweight convolution-attention and structural MRI, *Comput. Biol. Med.* (ISSN 0010-4825) 171 (2024) 108116, <https://doi.org/10.1016/j.compbiomed.2024.108116>.
- [42] Shakhnoza Muksimova, Sabina Umirzakova, Nargiza Iskhakova, Aziz Khaitov, Young Im Cho, Advanced convolutional neural network with attention mechanism for Alzheimer's disease classification using MRI, *Comput. Biol. Med.* (ISSN 0010-4825) 190 (2025) 110095, <https://doi.org/10.1016/j.compbiomed.2025.110095>.
- [43] Prashant Upadhyay, Pradeep Tomar, Satya Prakash Yadav, Advancements in Alzheimer's disease classification using deep learning frameworks for multimodal neuroimaging: a comprehensive review, *Comput. Electr. Eng.* (ISSN 0045-7906) 120, Part C (2024) 109796, <https://doi.org/10.1016/j.compeleceng.2024.109796>.
- [44] Shuyu Liu, Jingjing Zhou, Xuequan Zhu, Ya Zhang, Xinzhu Zhou, Shaoting Zhang, Zhi Yang, Ziji Wang, Ruoxi Wang, Yizhe Yuan, Xin Fang, Xiongying Chen, Yanfeng Wang, Ling Zhang, Gang Wang, Cheng Jin, An objective quantitative diagnosis of depression using a local-to-global multimodal fusion graph neural network, *Patterns* (ISSN 2666-3899) 5 (12) (2024) 101081, <https://doi.org/10.1016/j.patter.2024.101081>.